WEB PLATFORM

How Browsers Render Visual Guide

HTML parsing, CSSOM, the render tree,
layout, paint, and composite — pixels from markup.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Step 1: HTML Parsing -> DOM

Bytes become tokens, tokens become a tree of nodes.

PARSER BLOCKING

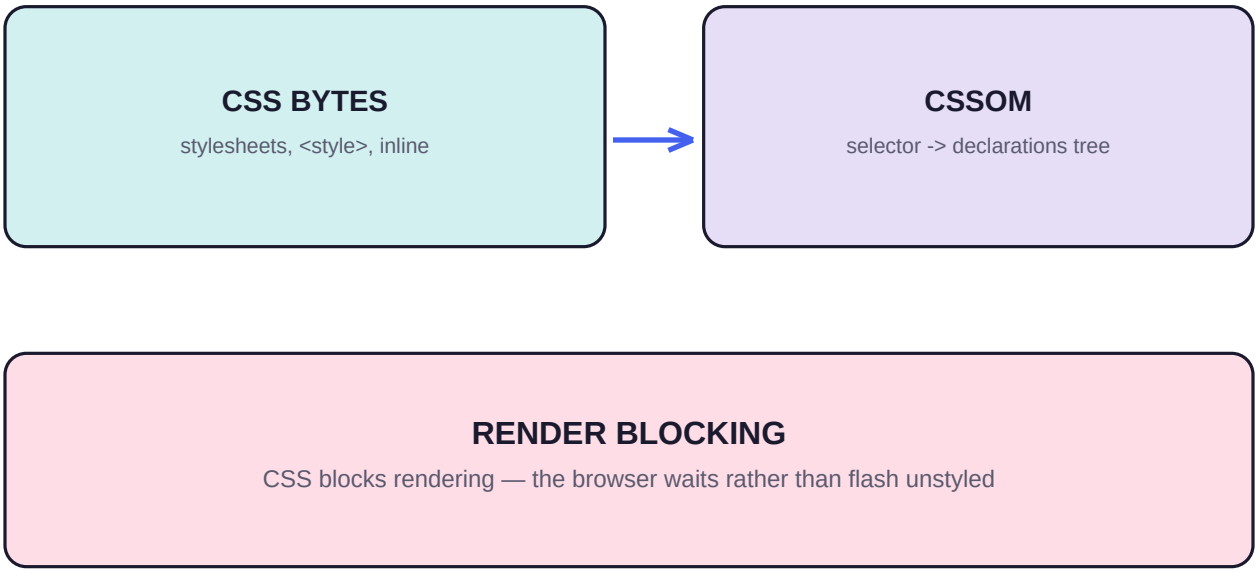
<script> without async/defer halts parsing until it runs

The DOM is the live, scriptable representation — what the HTML MEANS, not what it said.

STEP 2/6

Step 2: CSS -> CSSOM

Stylesheets become a second tree: every node's computed styles.

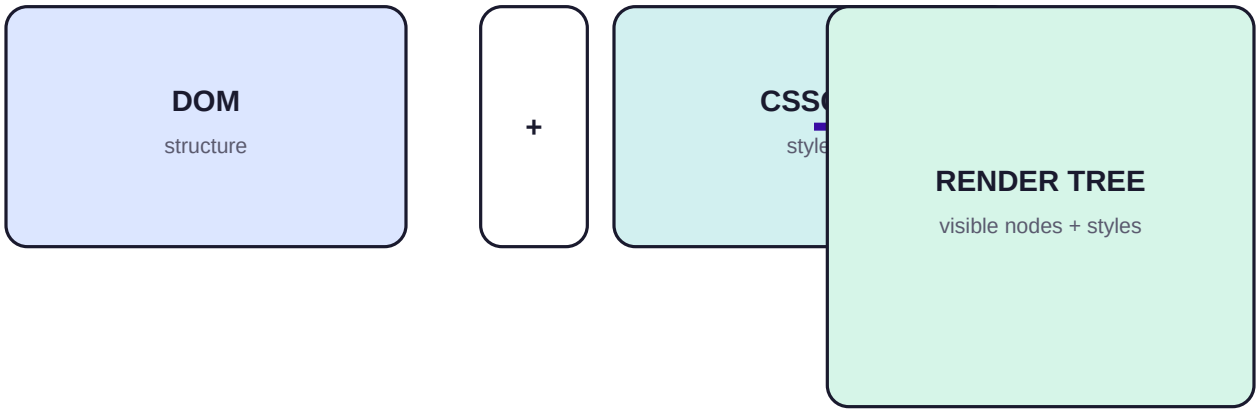


Inline critical CSS, defer the rest: the single biggest first-paint win on most sites.

STEP 3/6

Step 3: DOM + CSSOM = Render Tree

Only visible nodes, each carrying its computed style.



display:none and <head> nodes are dropped — they render nothing.

visibility:hidden stays (takes space); display:none is gone from the tree entirely.

STEP 4/6

Step 4: Layout -> Paint -> Composite

Three phases turn the tree into pixels.

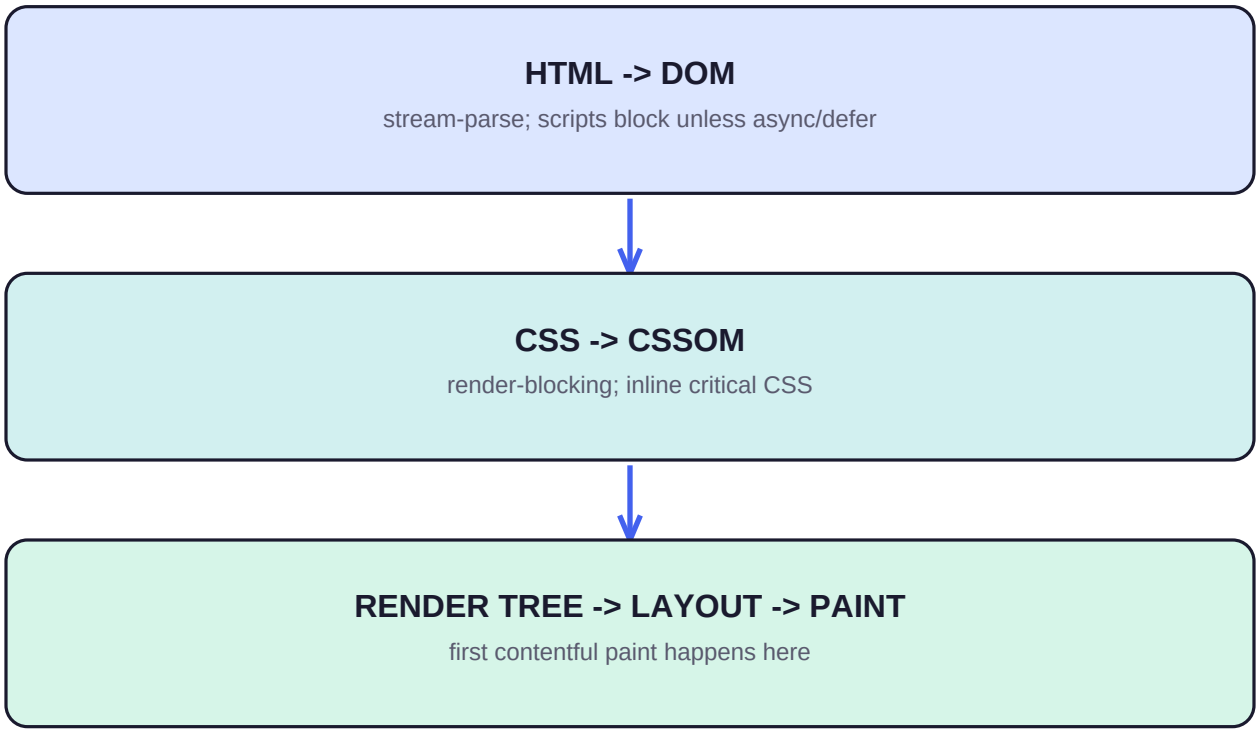
COST LADDER

layout > paint > composite — animate transform/opacity to stay in composite

STEP 5/6

Step 5: The Critical Rendering Path

The minimum work before first pixels: optimize this chain.



FCP, LCP, CLS: the Core Web Vitals measure this pipeline's speed and stability.

STEP 6/6

Rendering Performance

Reflow is expensive; repaint is cheaper; composite is cheapest.

REFLOW

geometry change — width, font, DOM insert

REPAINT

visual change — color, shadow (no geometry)

COMPOSITE-ONLY

transform + opacity animate on the GPU — 60fps territory

will-change / layers

promote sparingly — too many layers exhaust GPU memory

STEP ✓I✓

Key Takeaways

Browser Rendering — Remember

- HTML parses into the DOM tree; scripts block parsing unless async/defer.
- CSS builds the CSSOM — render-blocking by design to avoid unstyled flashes.
- Render tree = visible DOM nodes plus computed styles; display:none is excluded.
- Layout computes geometry, paint fills pixels, composite merges layers on the GPU.
- The critical path is HTML->DOM + CSS->CSSOM -> render tree -> first paint; shorten it.
- Animate transform and opacity to stay in the cheap composite phase.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev